

Mathematical Analysis of Time-series & Meteorological Factors in Electricity Demand Forecasting Based on Carbon Neutrality

Bo Jie

Graduate School of Engineering
The University of Tokyo
Tokyo, Japan
jiebo@asc.t.u-tokyo.ac.jp

Jumpei Baba

Graduate School of Frontier Sciences
The University of Tokyo
Tokyo, Japan
baba@asc.t.u-tokyo.ac.jp

Akiko Kumada

Graduate School of Engineering
The University of Tokyo
Tokyo, Japan
kumada@hvg.t.u-tokyo.ac.jp

Abstract—Due to the modernization of the electric power system and the increase of renewable energy, there is a significant need for various energy management systems (EMSs), in which electricity demand forecasting technology plays a remarkable role. Moreover, the load dispatching center operator regulates the supply-demand balance in the service area to meet the ever-changing electricity demand based on the demand forecasting results. Unfortunately, climatic and natural disaster factors bring uncertainties in the demand forecast process, which leads to the low prediction accuracy of the existing forecast techniques with failing to fully consider these factors affecting load fluctuations. In this paper, the importance of major time-series, meteorological and business-categorical energy consumption influencing factors applied to electricity demand forecasting is thus comprehensively and systematically illustrated through the correlation and regression analysis methods based on the real electricity demand data and various influencing factors data affecting electricity demand variation. The simulation results analysis in this study can assist load forecasters in more rationally identifying the influencing factors of maximum electricity consumption at various periods, thereby improving the overall accuracy of load forecasting.

Index Terms—electricity demand forecasting, correlation and regression analysis methods, time-series influencing factor, meteorological influencing factor, influence evaluation, forecasting accuracy

I. INTRODUCTION

In order to achieve carbon neutrality, the electrification in the energy consumption demand side is evolving in all societal sectors, and the demand for electricity consumption is becoming more complex and diverse. For example, expanding utilization of electric vehicles in the transportation sector, manufacturing processes automation and hydrogen production in the industrial sector, the introduction of solar panel and battery systems in electric power systems, etc. As Japanese Carbon Neutrality concept shown in Fig.1, multiple sectors electrification innovation is needed to achieve Carbon Neutrality by 2050 [1]. Due to the non-storability of electric energy and the variety and complexity of electric energy consumption loads, it is extremely vital for transmission and distribution system operators (TSO/DSO) to accurately forecast future electricity consumption demand. And the electricity Power generation plans can be scheduled more effectively.

In this line of work, there are various applications of prediction algorithms based on computational intelligence to forecast the electricity demand of an industrial pole for the next hour [2] [3]. For example, extensively employed machine learning based approach was applied to building energy systems [4]. Deep learning base approach, artificial neural network (ANN) [5], and long short-term memory (LSTM) prediction methods were analytically compared based on an empirical model decomposition model [6]. Moreover, two-stage integrated price and load forecasting framework [7] and prediction intervals estimations by using a differential evolution-based approach [8] were developed for day-ahead forecasting. Hybrid forecasting model based on improved particle swarm optimization (PSO) algorithm was employed to forecast electricity load demand [9]. Prediction errors reduction performance was demonstrated based on a smart energy forecasting architecture [10]. For choosing the least cost electric load forecasting model, root mean squared, mean absolute, and mean absolute percentage errors were considered simultaneously to correlate meteorological parameters [11]. However, bulk of studies on electricity load forecasting concentrate on anticipating a single time series that depicts the electricity load consolidated over a sizable number of electricity network nodes. Such as, Lahouar *et al.* proposed a model by using expert input selection method to predict the electrical load demand of day-ahead by a step of one hour in [12]. Amjady *et al.* performed a hybrid method with neural network, evolutionary algorithm and wavelet transform for short-term load forecasting with focusing on the hourly load forecasting [13]. Zhang *et al.* applied forecasting accuracy as an optimization objective function, the specific ratio value combinations of each characteristic parameter affecting the hourly electricity load as the input features of the forecasting model [14]. Semero *et al.* applied micro-grid hourly average load demand data in a hybrid approach for short-term load forecast [15]. Australian Energy Market Operator (AEMO) used a semi-parametric additive model to forecast half-hourly electricity demand [16]. Ziel *et al.* applied a prediction approach based on smoothed Bernstein online aggregation on individual load forecasting models

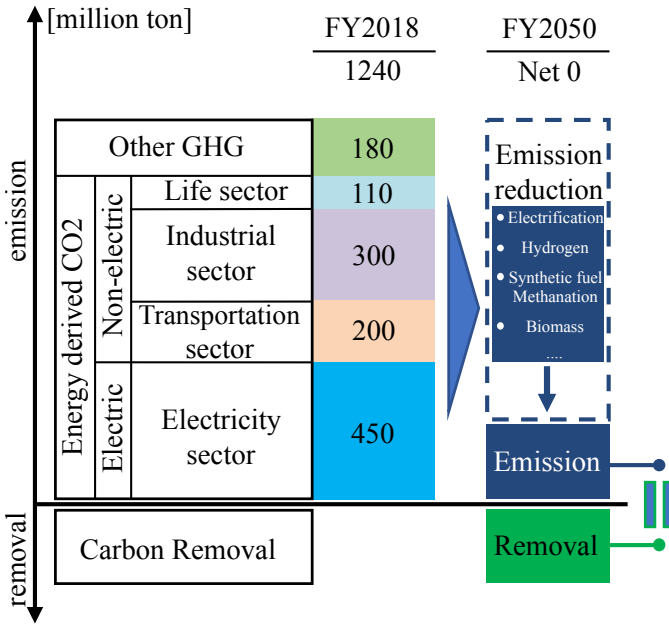


Fig. 1. Japanese Carbon Neutrality Concept. (created based on the information from [20])

[17]. Theodorakos *et al.* applied decomposition residuals deep neural networks (DR-DNN) to the short-term load forecasting problem [18]. Hyndman *et al.* forecasted the density of long-term peak electricity demand by using a fitted statistical model [19].

Electricity demand forecasting methods often apply machine learning on manually selected data that has influence on the electricity demand fluctuation. However, simply increasing the number of explanatory variables such as weather factors used for electricity demand forecasting does not necessarily improve forecast accuracy. Moreover, the so-called over-learning of machine learning can actually worsen the accuracy of the forecast. To avoid such problems, it is necessary to understand strict correlations when learning relationships from data. It can also have the effect of reducing the number of dimensions of explanatory variables used in electricity demand forecasting. The remaining part of this paper is consist of as follow: section II presents theoretical analysis on correlations and multiple regression analysis, section III illustrates influencing factors in electricity demand forecasting, section IV shows the overall correlation calculation results and analysis, and the conclusion and the future work land on the final section V.

II. CORRELATION & REGRESSION ANALYSIS METHODS

Influencing factors affecting electricity demand are complex and diverse, including weather conditions, economic conditions, and social factors (e.g., COVID-19, World Cup, etc.). Moreover, the magnitude of influence varies depending on the season, historical trends, and other variables. In general, when influencing factors are taken into account, mathematical analysis is essential for the data to be used in electricity

demand forecasting. Methods using correlation analysis and multiple regression analysis are widely employed.

A. Correlation Analysis

For decarbonization, in the industrial, commercial, residential, and transport sectors, in addition to improving energy consumption efficiency by thorough energy efficiency measures, promotion of electrification by decarbonized power sources is strongly advocated, and some concrete measures are also actively moving forward. For example, the heat demand and manufacturing process reform in industrial sector, the introduction of electric vehicle (EV) to replace fossil fuel vehicle in transport sector, and the electrification of household items in residential sector. These changes directly affect the electricity demand with more sophisticated variations, which leads to new influencing factors in electricity demand variations. The correlation relationship of variations of new influencing factors and electricity demand variations needs to be newly analyzed based on a large amount of accumulated data. And the correlation analysis method with Eqs. (1) ~ (4) shown below are often applied.

$$R_{D,F_x} = \frac{\text{cov}(D, F_x)}{\sqrt{\text{var}(D) \text{var}(F_x)}} \quad (1)$$

$$\begin{aligned} \text{cov}(D, F_x) &= E[(D - E[D])(F_x - E[F_x])] \\ &= E[DF_x] - 2E[D]E[F_x] + E[D]E[F_x] \\ &= E[DF_x] - E[D]E[F_x] \end{aligned} \quad (2)$$

$$\begin{cases} E[D] = \sum_{t=1}^n D_t P(D_t) \{D = D_t\} \\ E[F_x] = \sum_{t=1}^n F_{x,t} P(F_{x,t}) \{F_x = F_{x,t}\} \\ E[DF_x] = \sum_{t=1}^n \sum_{t=1}^n (D_t F_{x,t}) P(D_t F_{x,t}) \{D = D_t, F_x = F_{x,t}\} \end{cases} \quad (3)$$

$$\begin{cases} \text{var}(D) = \frac{\sum_{t=1}^n (D_t - \frac{1}{n} \sum_{t=1}^n D_t)^2}{n} \\ \text{var}(F_x) = \frac{\sum_{t=1}^n (F_{x,t} - \frac{1}{n} \sum_{t=1}^n F_{x,t})^2}{n} \end{cases} \quad (4)$$

D : electricity demand

F_x : influencing factor x (atmospheric temperature, precipitation, etc.)

E : mathematical expectation value

P : probability when ($D = D_t, F_x = F_{x,t}$)

t : time point of D & F_x

n : number of data of D & F_x

The value of variable R_{D,F_x} can be used to figure out the proper correlation relationship between the electricity demand and a certain influencing factor by the computation based on a large number of data:

- $R_{D,F_x} > 0$: positive correlation
- $R_{D,F_x} < 0$: negative correlation
- $R_{D,F_x} = 0$: no correlation.

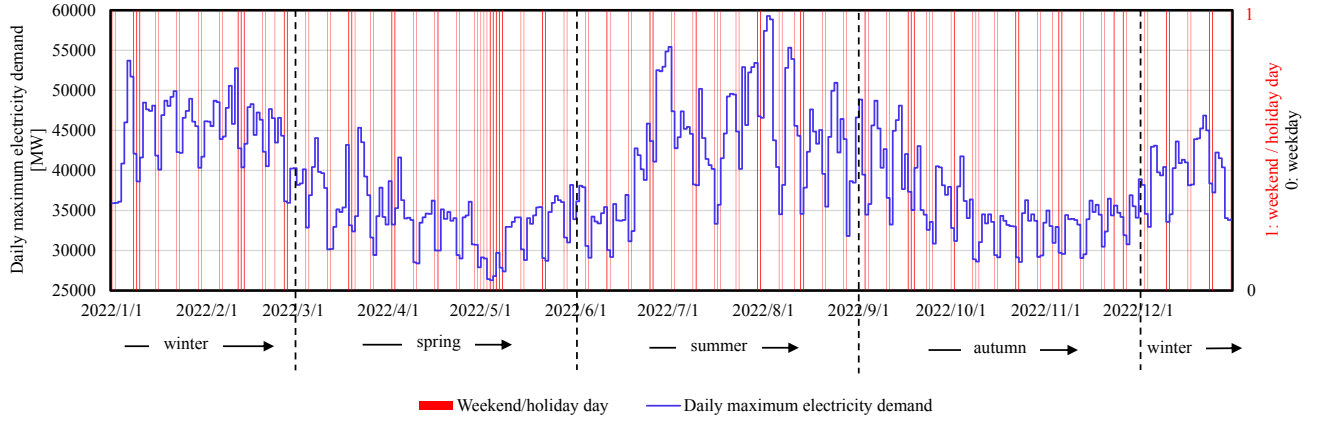


Fig. 2. Daily maximum electricity demand.

B. Multiple Regression Analysis

Multiple regression analysis is a statistical method in which the objective variable is expressed using multiple explanatory variables. In order to make forecasts using multiple regression analysis, it is necessary to examine factors that affect changes in electricity demand and select them as explanatory variables. Although there are countless candidates for explanatory variables, electricity demand is forecasted based on multiple regression equation. Eq.(5) expresses the influencing factor F_x that are correlated with electricity demand D as explanatory variables, and the regression parameters (ω_0 , ω_x) are calculated by Eqs.(6) and (7) based on historical data. As explanatory variables, weather data such as temperature, humidity, precipitation, as well as data on the day of the week, are analyzed in the following section.

$$\begin{cases} D = \omega_0 + \sum_{x=1}^m \omega_x F_x & (linear) \\ D = \omega_0 + \sum_{x=1}^m \omega_x (F_x)^x & (nonlinear) \end{cases} \quad (5)$$

$$L = \sum_{t=1}^n (D_t - \hat{D}_t)^2 \quad (6)$$

$$\frac{\partial}{\partial \omega_x} L = 0 \quad (7)$$

$\hat{D}_t$: electricity demand forecast value by Eq.(5) at time t

D_t : electricity demand real value at time t

ω_0 : forecasting bias

ω_x : weight of influencing factor x

m : the number of influencing factors ($x \in [1, m]$)

Climatic data such as the lowest temperature, the highest temperature as well as humidity are mainly employed as explanatory variables. Moreover, weather data such as sunshine and rain, and the information on the day of the week as non-weather variables also are quantified and employed. Explanatory variables that have a nonlinear relationship with electricity demand, such as temperature, may be expressed in a polynomial form by adding higher-order terms.

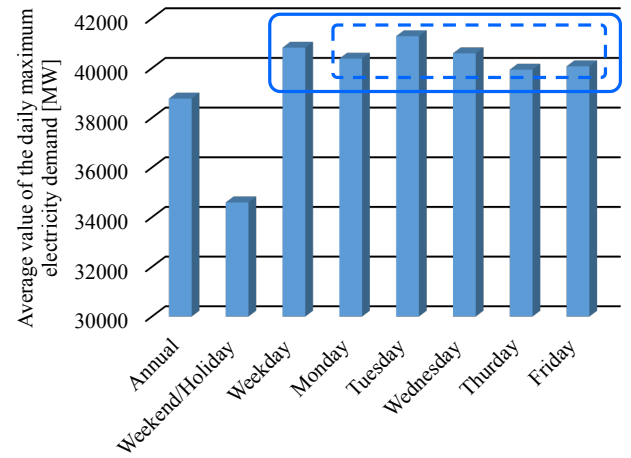


Fig. 3. Weekday analysis.

III. CORRELATION BETWEEN METEOROLOGICAL DATA & ELECTRICITY DEMAND

Electricity demand data in TEPCO's service area and meteorological data in the Tokyo area (JMA) are utilized in this study (data in hourly intervals) [21] [22]. The data is available from January 1st, 2022 to December 31st, 2022. As shown in Figure 3, the daily maximum electricity demand values are depicted as blue curves.

Seasonality: The daily maximum electricity demand curve in Fig.2 clearly indicates that electricity demand is low in the spring and fall seasons and high in the summer and winter seasons, which indicates that temperature has a relatively large impact on electricity demand in the Tokyo area.

Weekday-weekend/holiday: The red curve in Fig.2 indicates that the day is a weekend or a national holiday, whereas the rest are weekdays. Fig.3 depicts the yearly average of the maximum daily electricity demand for each day of the weekday and weekend/holiday throughout the whole year of 2022. Referring to Figs.2 and 3, it is evident that there is a significant difference between the electricity demand on weekdays and weekends/holiday: the electricity demand

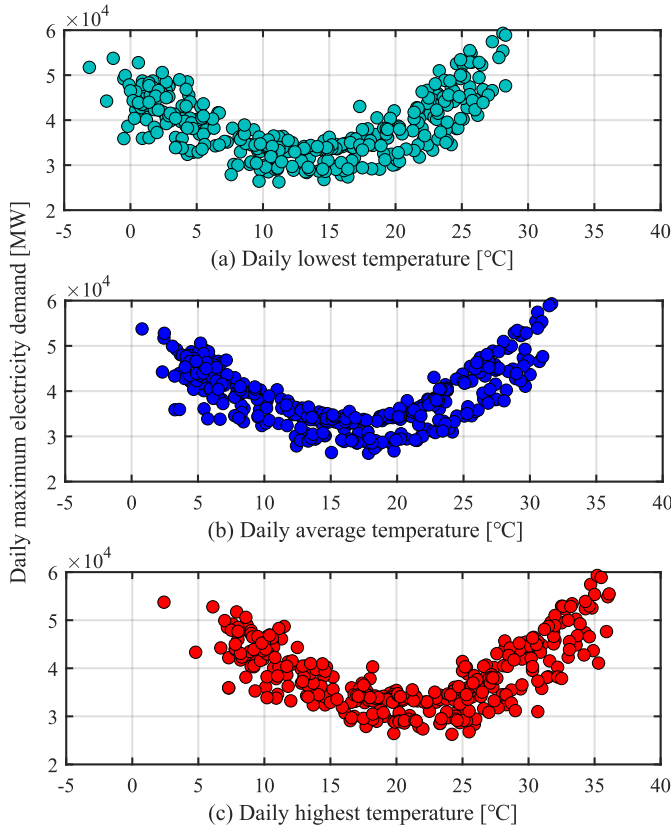


Fig. 4. Temperature analysis.

on weekend/holiday is significantly lower compared to the electricity demand on weekdays during the week; and the variation in electricity demand between weekdays is less pronounced than it is on weekend/holiday. Therefore, when considering the impact of weekday and weekend/holiday on the variation of electricity demand, it can be divided into two categories: one for weekday and one for weekend/holiday. In this study, as one of the factors influencing daily electricity demand fluctuation, weekend/holiday is quantified as “1” and weekdays are quantified as “0”.

Temperature: The correlation between the daily maximum electricity demand and the daily lowest temperature, the daily average temperature, and the daily highest temperature throughout one year is illustrated in Fig.4, which depicts a non-linear relationship between the daily maximum electricity demand and the daily lowest, average, and highest temperatures (2nd order function relationship). The result also coincides with the fact that the electricity demand rises when the temperature is very low (in winter) or very high (in summer). The daily electricity demand is directly and significantly related to the Air Conditioning heating/cooling operation.

Social sectors: Table I shows the total energy consumption of each social sector in the TEPCO service Area in 2022. The business/services and manufacturing sectors account for the majority of energy consumption. The main electricity consumption in the business/services sector is lighting and

TABLE I
SECTORAL ENERGY CONSUMPTION IN TEPCO SERVICE AREA IN 2022
(TJ: TERAJOULE).

Sector	Energy consumption [TJ]	Percentage
Agriculture/Forestry	2388	0.16%
Fishing	2148	0.15%
Mining/Quarrying	3176	0.22%
Construction	40346	2.78%
Manufacturing	467469	32.19%
Business/Service	936584	64.50%
Total	1452111	100.00%

heating/cooling.

IV. RESULTS & ANALYSIS

From the aforementioned study, a fact can be obtained that the electricity demand fluctuation is affected by the life and working patterns of residents living in the Tokyo area. The life and working pattern of the residents is directly related to the variations in weather conditions. Therefore, in this article, calculations is performed mainly for the magnitude of influence of weather influencing factors on the electricity demand fluctuation. The results were based on the calculation of data for fore seasons in 2022 in Tokyo area. As shown in Table II and Fig.5.

A. Results

In Fig.5, the relationship between five meteorological indicators and daily maximum electricity demand is presented for four seasons in 2022 in Tokyo area. The effect of precipitation on the daily maximum electricity demand fluctuation is not significant. The effect of humidity on the daily maximum electricity demand fluctuation is not significant in the spring, autumn and winter. In summer, electricity demand fluctuation tends to decrease as air humidity increases, due to the fact that as air humidity increases, air temperature decreases to a certain extent, the needs of electricity consumption due to air conditioning cooling decreases. The effect of the air temperature variation on the daily maximum electricity demand is different in the four seasons. In this study, daily lowest air temperature, daily average air temperature, and daily highest air temperature were focused on. The lowest temperatures are in the late nighttime period, the demand for electricity consumption in late nighttime period is the lowest in the day and not the maximum daily electricity demand. In contrast, the average and maximum daily temperatures have an impact on the daily maximum electricity consumption and show different correlations in the four seasons. In summer and winter, the linear relationship is more obvious, in summer, as the temperature increases, the electricity demand increases with it, and in winter the opposite tendency is observed, due to the variation in electricity consumption for air conditioning. In spring and autumn, the trend of the low temperature segment is the same as that of winter, due to the use of air

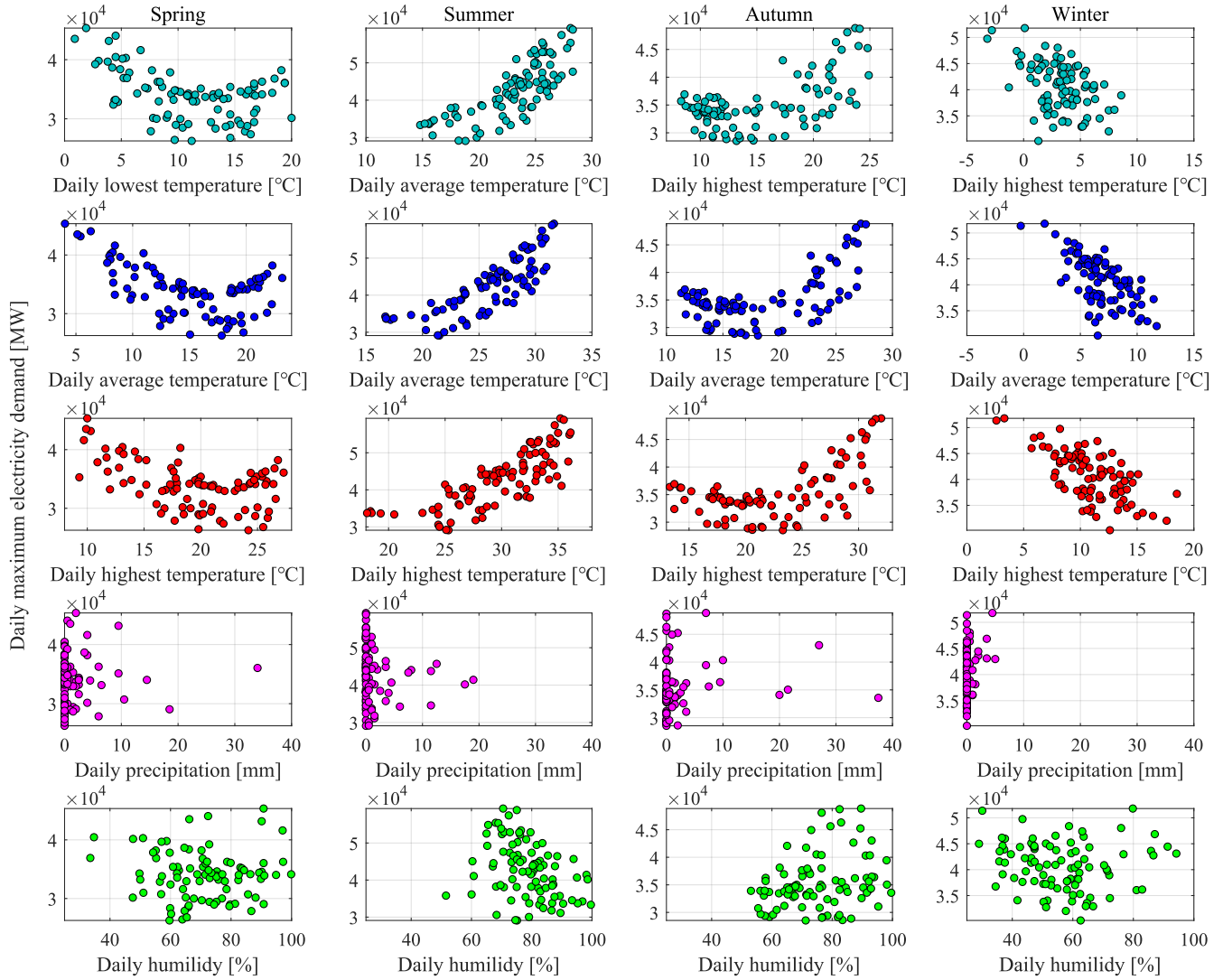


Fig. 5. Daily maximum electricity demand based on 4 seasons in Tokyo area.

conditioning heaters in the low temperature spring, whereas the trend of the high temperature segment is the same as that of summer. The magnitude of the influence of meteorological factors on the daily maximum electricity demand is shown in Table II. Negative correlations are indicated by negative values, positive correlations are indicated by positive values, and higher absolute values indicate a higher magnitude of influence.

B. Extensible Analysis

The consumption of electricity is strongly related to social and economic activities, and has the characteristic of repeatedly changing with the daily, weekly, seasonal, and other cycles. On a daily unit basis, electricity consumption is high during the daytime when economic activities are active and people are in numerous moving about, and it is lowest during the late-night hours when there is little activity. On a weekly basis, the amount of electricity used is higher during the daytime on weekdays and higher during the evening

TABLE II
INFLUENCING COEFFICIENT OF METEOROLOGICAL FACTORS ON DAILY ELECTRICITY DEMAND FLUCTUATION.

Daily value	Spring	Summer	Autumn	Winter
Lowest temperature	-0.47	0.78	0.59	-0.44
Average temperature	-0.53	0.81	0.61	-0.63
Highest temperature	-0.52	0.88	0.56	-0.65
Precipitation	0.036	-0.11	0.12	0.26
Humidity	-0.03	-0.35	0.27	-0.09

and nighttime on weekends than during the daytime, but the amount of electricity consumed per day is lower than that on weekdays. In terms of seasonal units, the consumption is higher in summer and winter due to cooling demand and heating demand, respectively, however, it is lower in spring and fall, which are the transitional periods between the two periods. In order to ensure a stable supply of electricity in

response to daily, weekly, and seasonal fluctuations in demand, electric power companies prepare annual, monthly, weekly, and daily operation plans for nuclear, thermal, and hydro-electric power generators, as well as daily operation plans for natural renewable power sources, which are superior in terms of environmental friendliness and energy self-sufficiency, and the amount of supply expected in the operation plan.

In order to meet the challenges facing future electric power systems, the needs for accurate demand forecasting increase, and the needs themselves also diversify. For example, as the share of renewable energy sources increases, there is an even greater need for more accurate forecasting of the residual demand that should be provided by conventional power generation and other sources. Negawatt transactions for balancing supply and demand requires the forecast of the responsiveness during demand response. In addition, as electric vehicles (EVs) become more widespread, it will be necessary to forecast the charging demand of electric vehicles (EVs), which will have an influence on electricity demand.

Moreover, various energy management systems (EMSs) have been aggressively implemented on electricity demand side in an effort to minimize the electricity consumption and rationalize equipment operation schedules. In EMS, load forecasting for the short-term future period is helpful for power consumers to regulate manufacturing plans in advance to reduce energy consumption. Forecasting technology for power consumption on the demand side itself is becoming crucially influential. The factors that influence the fluctuation of power consumption on the demand side have a higher diversity and uncertainty. And the correlation analysis of these influencing factors is furthermore crucial and vital.

In order to meet the diverse needs for demand forecasting required in future electric power systems, a concept that flexibly combines the machine learning with demand big data obtained from smart meters and weather big data obtained from numerical weather forecast models, weather satellite images, and so forth, is required.

V. CONCLUSION

Illustrative analysis based on real data of electricity demand and meteorological indexes is performed as a cut-off point for electricity demand forecast with a higher accuracy. Meteorological factors that affect electricity demand fluctuation, vary by region and season. The analysis is conducted on the weather characteristics of the electricity demand forecasting area and its relationship with the weather conditions. The results contribute to higher accuracy and efficiency in demand and supply management with power grid modernization for achieving carbon neutrality.

As future work, due to the electricity demand has various fluctuation trends and relationships with time-series, meteorological influencing factors depending on different voltage level composition, such as high voltage power grid, low voltage EMS. Electricity demand forecasting methods using machine learning is intend to be employed to conduct short-time period demand forecasts based on historical big data.

REFERENCES

- [1] Ministry of Economy, Trade and Industry (METI), Japan, "The 6th Strategic Energy Plan," [Available]: https://www.enecho.meti.go.jp/en/category/others/basic_plan/, (accessed 2022/11/29).
- [2] R. Porteiro, S. Nesmachnow, and L. Hernández-Callejo, "Short term load forecasting of industrial electricity using machine learning," in *Smart Cities*, S. Nesmachnow and L. Hernández Callejo, Eds. Cham: Springer International Publishing, 2020, pp. 146–161.
- [3] M. Farrokhhabadi, J. Browell, Y. Wang, S. Makonin, W. Su, and H. Zareipour, "Day-ahead electricity demand forecasting competition: Post-covid paradigm," *IEEE Open Access Journal of Power and Energy*, vol. 9, pp. 185–191, 2022.
- [4] A. T. Eseye, M. Lehtonen, T. Tukia, S. Uimonen, and R. John Millar, "Machine learning based integrated feature selection approach for improved electricity demand forecasting in decentralized energy systems," *IEEE Access*, vol. 7, pp. 91 463–91 475, 2019.
- [5] M. Charytoniuk, E. Box, W.-J. Lee, M.-S. Chen, P. Kotas, and P. Van Olinda, "Neural-network-based demand forecasting in a deregulated environment," *IEEE Transactions on Industry Applications*, vol. 36, no. 3, pp. 893–898, 2000.
- [6] J. Bedi and D. Toshniwal, "Empirical mode decomposition based deep learning for electricity demand forecasting," *IEEE Access*, vol. 6, pp. 49 144–49 156, 2018.
- [7] L. Wu and M. Shahidehpour, "A hybrid model for integrated day-ahead electricity price and load forecasting in smart grid," *IET Generation, Transmission & Distribution*, vol. 8, no. 12, pp. 1937–1950, 2014.
- [8] N. A. Shrivastava and B. K. Panigrahi, "Prediction interval estimations for electricity demands and prices: a multi-objective approach," *IET Generation, Transmission & Distribution*, vol. 9, no. 5, pp. 494–502, 2015.
- [9] Y. He, F. Guang, and R. Chen, "Prediction of electricity demand of china based on the analysis of decoupling and driving force," *IET Generation, Transmission & Distribution*, vol. 12, no. 13, pp. 3375–3382, 2018.
- [10] Y. Cheng, C. Xu, D. Mashima, P. P. Biswas, G. Chipurupalli, B. Zhou, and Y. Wu, "Powernet: a smart energy forecasting architecture based on neural networks," *IET Smart Cities*, vol. 2, no. 4, pp. 199–207, 2020.
- [11] M. Jawad, M. S. A. Nadeem, S.-O. Shim, I. R. Khan, A. Shaheen, N. Habib, L. Hussain, and W. Aziz, "Machine learning based cost effective electricity load forecasting model using correlated meteorological parameters," *IEEE Access*, vol. 8, pp. 146 847–146 864, 2020.
- [12] A. Lahouar and J. Ben Hadj Slama, "Day-ahead load forecast using random forest and expert input selection," *Energy Conversion and Management*, vol. 103, pp. 1040–1051, 2015.
- [13] N. Amjadi and F. Keynia, "Short-term load forecasting of power systems by combination of wavelet transform and neuro-evolutionary algorithm," *Energy*, vol. 34, no. 1, pp. 46–57, 2009.
- [14] G. Zhang and J. Guo, "A novel method for hourly electricity demand forecasting," *IEEE Transactions on Power Systems*, vol. 35, no. 2, pp. 1351–1363, 2020.
- [15] Y. K. Semero, J. Zhang, and D. Zheng, "Emd-pso-anfis-based hybrid approach for short-term load forecasting in microgrids," *IET Generation, Transmission & Distribution*, vol. 14, no. 3, pp. 470–475, 2020.
- [16] S. Fan and R. J. Hyndman, "Short-term load forecasting based on a semi-parametric additive model," *IEEE Transactions on Power Systems*, vol. 27, no. 1, pp. 134–141, 2012.
- [17] F. Ziel, "Smoothed bernstein online aggregation for short-term load forecasting in ieee dataport competition on day-ahead electricity demand forecasting: Post-covid paradigm," *IEEE Open Access Journal of Power and Energy*, vol. 9, pp. 202–212, 2022.
- [18] K. Theodorakos, O. M. Agudelo, M. Espinoza, and B. De Moor, "Decomposition-residuals neural networks: Hybrid system identification applied to electricity demand forecasting," *IEEE Open Access Journal of Power and Energy*, vol. 9, pp. 241–253, 2022.
- [19] R. J. Hyndman and S. Fan, "Density forecasting for long-term peak electricity demand," *IEEE Transactions on Power Systems*, vol. 25, no. 2, pp. 1142–1153, 2010.
- [20] Ministry of Environment, Japan, "National Greenhouse Gas Inventory Report of Japan (NIR)," [Available]: <https://www.nies.go.jp/gio/en/ab/outghg/index.html>, (accessed 2023/03/01).
- [21] Tokyo Electric Power Company Holdings, [URL]: <https://www.tepco.co.jp/en/hd/index-e.html>.
- [22] Japan Meteorological Agency, [URL]: <https://www.jma.go.jp/jma/indexe.html>.